

MORNINGTON ISLAND, AustraliaWMO: **942560**Lat: **16.664S**Lon: **139.184E**Elev: **29**StdP: **14.68**Time Zone: **10.00 (E10)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	57.3	59.9	38.3	34.1	74.3	41.4	38.4	71.9	20.6	65.1	18.9	66.3	2.5	140	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.1	93.9	75.1	92.5	76.4	91.3	77.1	83.2	88.3	82.3	87.5	81.5	86.8	8.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.9	166.2	86.1	80.9	160.9	85.6	80.0	156.1	84.9	47.3	88.4	46.3	87.5	45.3	87.0	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	16.1	14.4	52.3	99.1	2.8	1.7	50.3	100.3	48.7	101.3	47.1	102.2	45.1	103.4
			48.1	83.0	2.0	1.8	46.7	84.3	45.5	85.3	44.3	86.3	42.9	87.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.2	83.3	83.6	82.4	81.5	76.3	71.3	70.5	72.5	78.0	82.2	84.5	84.9
	DBStd	6.06	3.08	2.61	2.62	2.99	3.87	4.54	3.63	3.69	3.00	2.74	2.64	2.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	11	0	0	0	0	0	6	3	2	0	0	0	0
	CDD50	10666	1033	940	1003	944	814	640	636	698	841	999	1036	1082
	CDD65	5200	568	520	538	494	349	195	174	234	391	534	586	617
	CDH74	54844	6834	6211	6022	5352	2716	971	781	1503	3294	5864	7325	7971
	CDH80	20880	2757	2468	2232	2015	698	150	87	287	1016	2224	3262	3684
Wind	WSAvg	6.7	6.7	5.4	5.4	5.7	6.9	7.7	7.2	6.7	7.1	7.2	7.4	6.7
Precipitation	PrecAvg	50.6	12.4	13.0	11.5	2.6	0.4	0.2	0.1	0.0	0.1	0.6	2.2	6.7
	PrecMax	83.2	34.7	51.0	26.2	19.2	5.0	3.1	0.9	0.2	1.4	6.6	8.9	18.1
	PrecMin	23.0	2.6	1.6	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8
	PrecStd	15.7	7.9	10.9	7.2	3.6	0.9	0.6	0.2	0.0	0.2	1.3	2.1	4.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	93.4	93.4	93.6	90.5	86.4	85.2	87.3	91.3	94.5	96.9	96.4
		MCWB	79.4	80.6	77.7	76.7	74.6	73.9	71.5	68.9	67.3	70.9	71.2	72.2
	2%	DB	91.4	91.4	91.5	91.2	87.7	83.6	82.2	84.9	88.5	91.7	93.7	93.4
		MCWB	80.0	80.6	77.7	76.8	74.2	71.8	69.0	68.2	69.9	73.1	73.8	75.9
	5%	DB	90.1	90.2	90.1	89.5	85.7	81.6	80.3	83.1	86.9	89.9	91.8	91.8
		MCWB	79.9	80.1	77.8	76.1	72.9	70.2	67.5	67.6	70.6	73.2	74.9	77.4
	10%	DB	88.9	89.0	88.5	88.1	84.0	79.4	78.4	81.4	85.4	88.5	90.2	90.2
		MCWB	79.6	79.8	77.7	75.7	72.1	68.5	66.4	66.9	70.5	73.6	75.7	77.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.5	85.6	83.5	81.7	78.8	76.8	73.5	74.2	77.0	81.3	83.3	81.5
		MCDB	89.4	90.8	87.9	87.1	85.6	82.9	81.2	81.9	83.5	86.9	89.8	89.7
	2%	WB	82.5	83.7	82.0	80.3	77.1	74.6	71.6	72.5	75.8	79.9	81.9	80.6
		MCDB	88.3	88.2	87.1	86.1	84.0	80.9	79.1	79.9	82.1	85.3	88.1	88.3
	5%	WB	81.6	82.6	81.0	79.2	75.5	72.7	70.1	71.4	74.8	78.6	80.0	79.9
		MCDB	87.5	87.1	86.0	85.0	82.6	79.0	77.3	78.4	81.1	84.4	86.6	87.5
	10%	WB	80.6	81.7	80.0	78.3	74.2	70.6	68.6	70.2	74.0	77.4	78.8	79.2
		MCDB	86.7	86.2	85.0	84.3	81.4	77.4	75.8	77.2	80.4	83.7	85.7	86.7
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.4	10.8	13.0	14.0	13.8	15.6	17.5	15.3	12.9	11.4	10.1
		MCDBR	10.1	11.2	13.3	14.7	15.8	16.4	17.7	19.0	16.3	14.6	13.5	11.7
		MCWBR	4.1	4.2	5.4	5.8	6.6	7.8	8.1	7.8	6.3	6.4	6.2	4.6
	5% WB	MCDBR	9.1	9.3	10.9	12.3	13.9	14.8	16.6	17.0	13.7	12.1	10.5	9.8
		MCWBR	4.5	4.2	5.7	5.8	6.6	7.6	8.2	8.0	5.6	5.4	5.4	4.3
Clear-Sky Solar Irradiance	taub	0.413	0.412	0.391	0.367	0.343	0.331	0.326	0.333	0.382	0.411	0.433	0.425	
	taud	2.532	2.534	2.589	2.630	2.640	2.630	2.633	2.592	2.459	2.407	2.396	2.461	
	Ebn at Noon	296	293	293	289	285	284	288	295	291	290	288	292	
	Edh at Noon	36	35	32	29	27	26	27	30	36	39	40	38	
All-Sky Solar Radiation	RadAvg	1709	1764	1775	1772	1594	1473	1572	1822	2104	2227	2143	1932	
	RadStd	218	259	209	152	85	86	39	40	60	110	141	212	

Historical Trends

	DBAvg	Heating
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